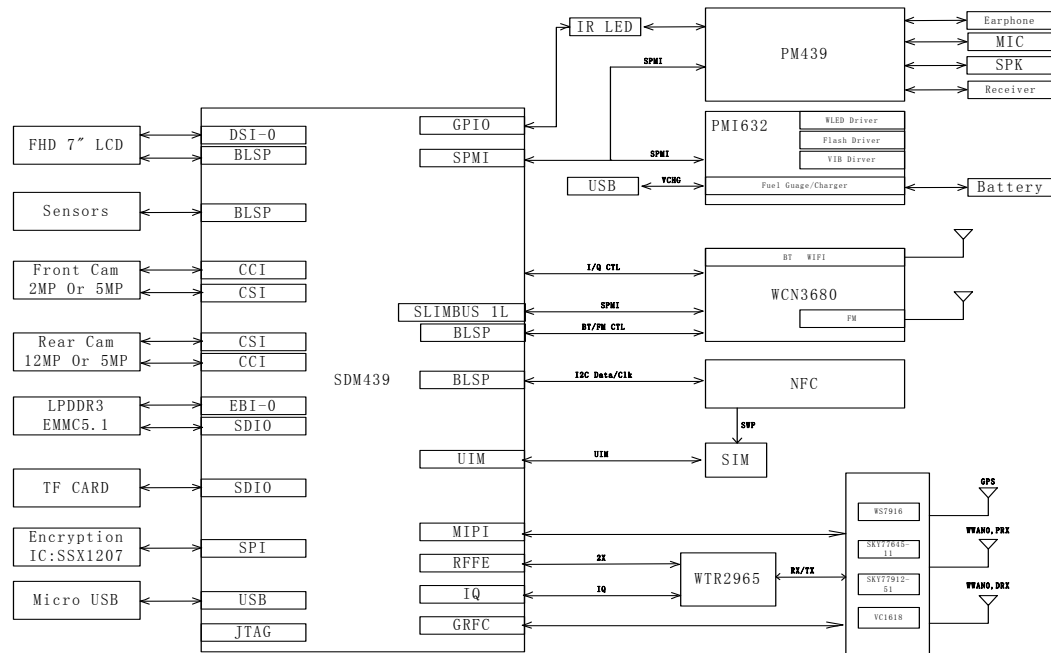


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Block
Diagram

SDM439 GPIO Configuration For QRD439+PMI632

GPIO_0	NFC_SPI_ESE_MOSI	GPIO_41		GPIO_82	FM_DATA
GPIO_1	NFC_SPI_ESE_MISO	GPIO_42	ACCL_INT1	GPIO_83	BT_CTL
GPIO_2	NFC_SPI_CS_N_0	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	NFC_SPI_ESE_CLK	GPIO_44	MAG_DRDY_INT	GPIO_85	FP_SPI_MOSI
GPIO_4	QUP_UART_TX_2	GPIO_45	GYRO_INT	GPIO_86	FP_SPI_MISO
GPIO_5	QUP_UART_RX_2	GPIO_46		GPIO_87	FP_SPI_CS
GPIO_6	WSA_SMB_I2C_SDA	GPIO_47	SENSOR_SPI_CS2_N	GPIO_88	FP_SPI_CLK
GPIO_7	WSA_SMB_I2C_SCL	GPIO_48	FP_INT_N1	GPIO_89	
GPIO_8	TP_SPI_MOSI	GPIO_49	UIM_BATT_ALARM	GPIO_90	
GPIO_9	TP_SPI_MISO	GPIO_50		GPIO_91	KEY_VOL_UP_N
GPIO_10	TP_SPI_CS_SDA	GPIO_51	UIM1_DATA	GPIO_92	
GPIO_11	TP_SPI_CLK_SCL	GPIO_52	UIM1_CLK	GPIO_93	NFC_ESE_PWR_REQ
GPIO_12	WSA_INTR	GPIO_53	UIM1_RESET	GPIO_94	WSA_IO_DATA
GPIO_13	LCD_ID	GPIO_54	UIM1_PRESENT	GPIO_95	WSA_IO_CLK
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	WSA_EN
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	HomeKey_FP_INT
GPIO_16	NFC_DISABLE	GPIO_57	UIM2_RESET	GPIO_98	
GPIO_17	NFC_IRQ	GPIO_58	UIM2_PRESENT	GPIO_99	
GPIO_18	NFC_I2C_SDA	GPIO_59		GPIO_100	RFFE1_CLK
GPIO_19	NFC_I2C_SCL	GPIO_60	LCD0_RESET_N	GPIO_101	RFFE1_DATA
GPIO_20	SENSOR_SPI_MOSI	GPIO_61	SMB_INT	GPIO_102	RFFE2_CLK
GPIO_21	SENSOR_SPI_MISO	GPIO_62		GPIO_103	RFFE2_DATA
GPIO_22	SENSOR_SPI_CS0_N	GPIO_63		GPIO_104	
GPIO_23	SENSOR_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	
GPIO_25	WSA_MCLK	GPIO_66		GPIO_107	GRFC3_SEL
GPIO_26	CAM_MCLK0	GPIO_67	SDCARD_DET_N	GPIO_108	
GPIO_27	CAM_MCLK1	GPIO_68		GPIO_109	GRFC5_SEL
GPIO_28		GPIO_69	CDC_PDM_CLK	GPIO_110	GRFC6_SEL
GPIO_29	CAM_I2C_SDA0	GPIO_70	CDC_PDM_SYNC	GPIO_111	GRFC7_SEL
GPIO_30	CAM_I2C_SCL0	GPIO_71	CDC_PDM_TX	GPIO_112	GRFC8_SEL
GPIO_31	CAM_I2C_SDA1	GPIO_72	CDC_PDM_RX0	GPIO_113	GRFC9_SEL
GPIO_32	CAM_I2C_SCL1	GPIO_73	CDC_PDM_RX1	GPIO_114	GRFC10_SEL
GPIO_33		GPIO_74	CDC_PDM_RX2	GPIO_115	
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSBI	GPIO_116	
GPIO_35	CAM_AVDD_LDO_EN	GPIO_76	WL_CMD_DATA_2	GPIO_117	
GPIO_36	MCAM_RST_N	GPIO_77	WL_CMD_DATA_1	GPIO_118	EXT_GPS_LNA_EN
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	CH0_GSM_TX_PHASE_D0
GPIO_38	SCAM_RST_N	GPIO_79	WL_CMD_SET	GPIO_120	RFFE5_CLK
GPIO_39		GPIO_80	WL_CMD_CLK	GPIO_121	RFFE5_DATA
GPIO_40		GPIO_81	FM_SSBI	GPIO_122	

GPIO_123			
GPIO_124	FP_RESET		
GPIO_125			
GPIO_126			
GPIO_127			
GPIO_128			
GPIO_129			
GPIO_130	NFC_DWL_REQ		
GPIO_131			
GPIO_132	RCM_MARKER1		
GPIO_133	RCM_MARKER2		

PMI632 GPIO/MPP Configuration

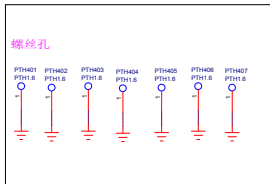
GPIO_1	CONNECTOR_THERMAL	GPIO_5	FLASH_STROBE_NOW
GPIO_2	SMB_PARALLEL_CHG_EN	GPIO_6	NEBULA_PWM
GPIO_3	SKIN_THEREMAL	GPIO_7	SMB_VCHG_P
GPIO_4	SMB_THEREMAL	GPIO_8	SMB_VCHG_M

PM439 GPIO/MPP Configuration

GPIO_1		MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	NFC_CLK_REQ		
GPIO_3	UIM_BATT_ALARM	MPP_3	VREF_DAC_MPP_3
GPIO_4	WLED_EN	MPP_4	QUIET_THERM_TP_LED_K
GPIO_5			
GPIO_6			
GPIO_7			
GPIO_8	LCM_BL_PWM		

GPIO
TABLE

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Doc ID	1200-PP432	Rev A								
Date	1/20/2018/18	Sheet 3 of 30								



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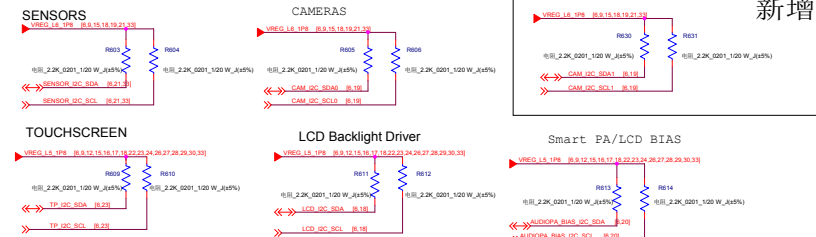


GPIO_01	GPIO01_GPIO_0000
GPIO_02	GPIO02_GPIO_0000
GPIO_03	GPIO03_GPIO_0000
GPIO_04	GPIO04_GPIO_0000
GPIO_05	GPIO05_GPIO_0000
GPIO_06	GPIO06_GPIO_0000

GPIO_07	GPIO07_GPIO_0000
GPIO_08	GPIO08_GPIO_0000
GPIO_09	GPIO09_GPIO_0000
GPIO_10	GPIO10_GPIO_0000
GPIO_11	GPIO11_GPIO_0000
GPIO_12	GPIO12_GPIO_0000

Default Boot Config (GPIO0) is GPIO0(GPIO0)

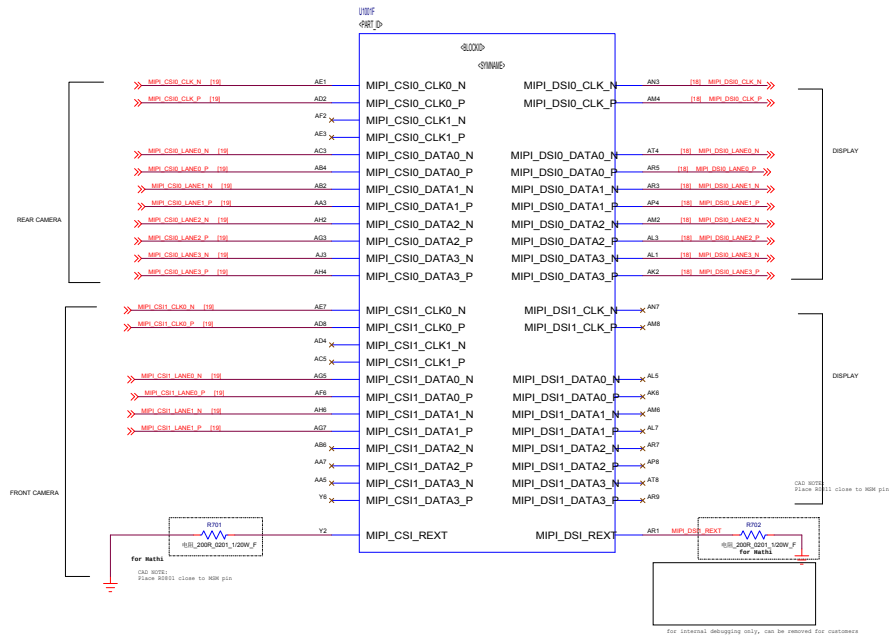
I2C PULL-UP RESISTORS



NOTE:
Ensure SW sets these GPIOs (Sensor, CTF and Camera I2C bus) to inout pull down when the peripherals are powered off to eliminate leakage.

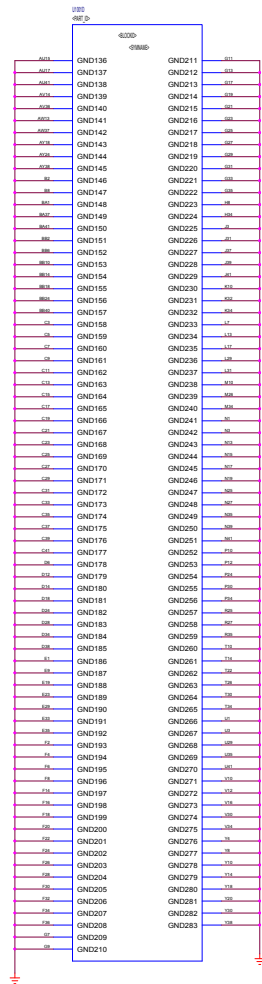
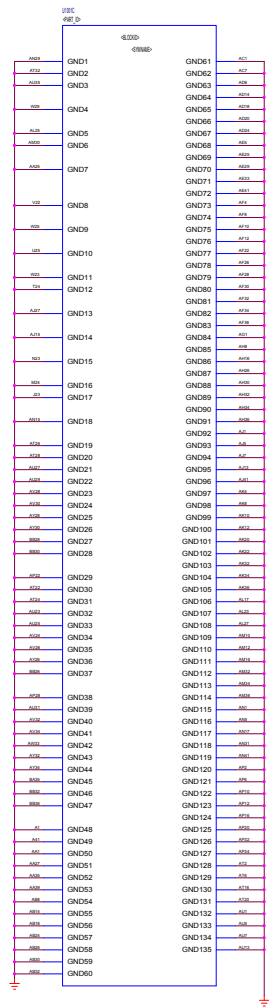
SDM439 GPIO

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Part Number		SDM439		Revision	
Version		1.0.0		Date	
Author		Qualcomm		Page	
Drawn By		Qualcomm		Page	



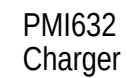
SDM439 MIPI

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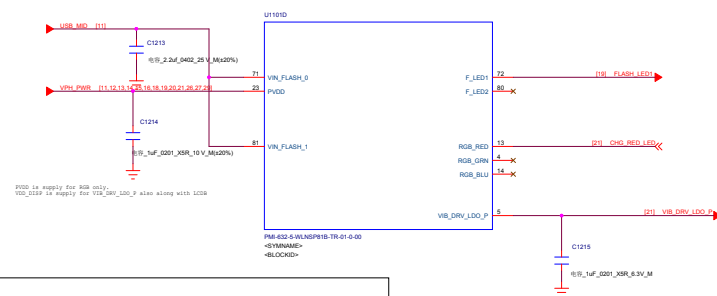
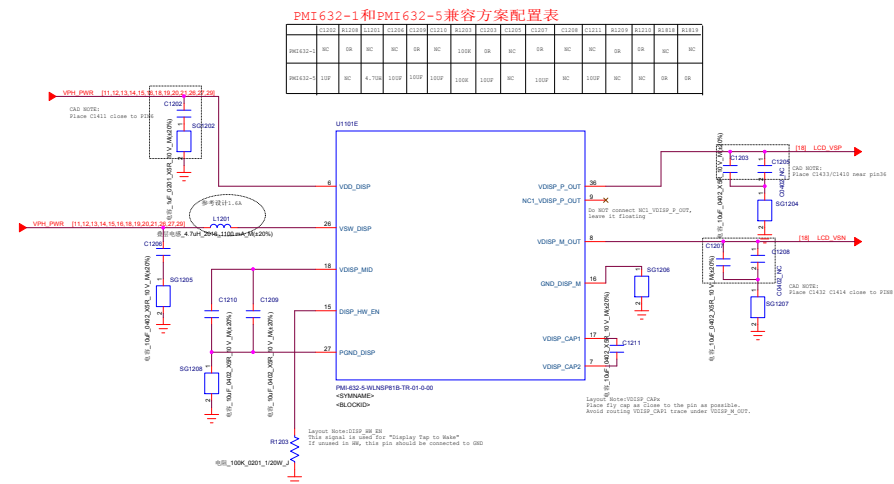
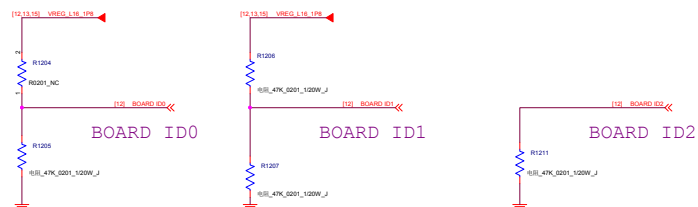
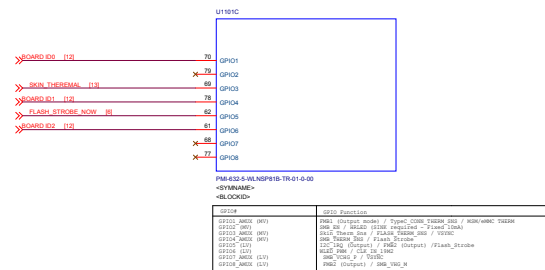
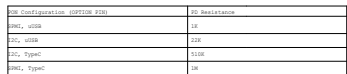


SDM439 GND

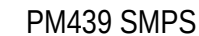
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PMI632
Control/Interface

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Date	Friday, March 08, 2019	Sheet	14 of 33

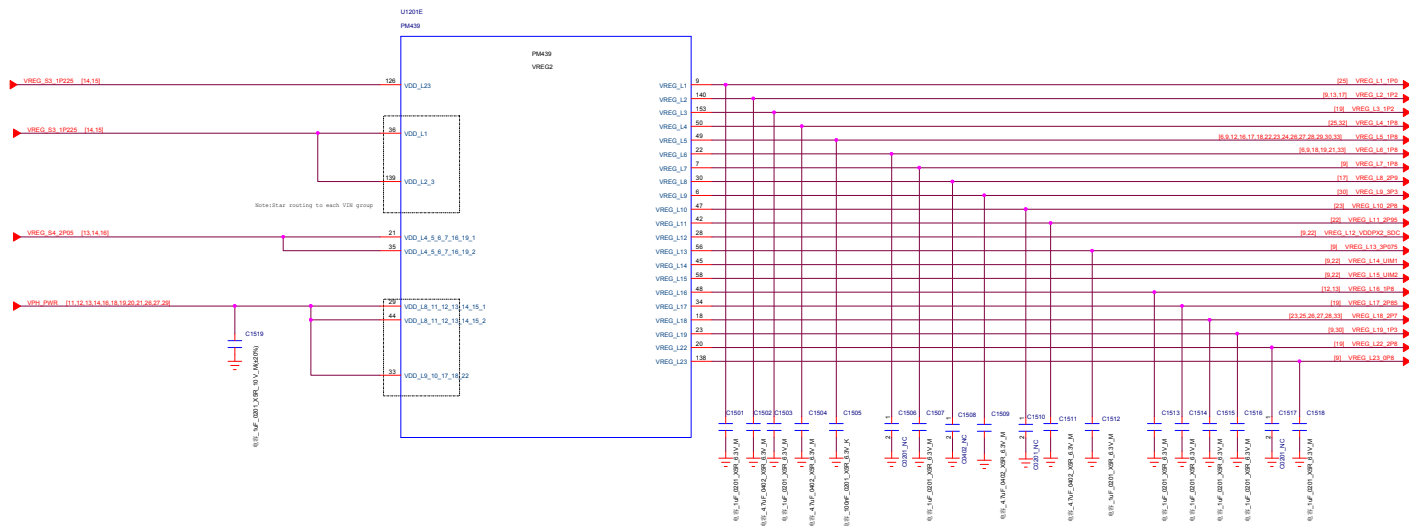


Table 3-6 PM439 regulators and their intended uses for SOM439/SOM429 when paired with PM632

Function	Circuit type	Default voltage (V) ¹	Specified range (V) ¹ (SOM439/SOM429)	Programmable range (V)	Rated current (mA)	Default on	Expected use (SOM439/SOM429)
S1	SMPS	0.744	0.4-1.14	0.33-2.04	3000	N	MSM modem
S2	SMPS	0.744	0.4-1.14	0.33-2.04	4000	Y	MSM core and graphics
S3	SMPS	1.28	1.2-1.25	0.33-2.04	2000	Y	MPI CSI and DSI. Low-voltage LDOs (1, 2, 3, and 23)
S4	SMPS	2.04	1.8-2.04	0.33-2.04	2000	Y	High-voltage LDOs (4, 5, 6, 7, 16, and 19)
S5	SMPS	0.735	0.4-1.14	0.350-1.355	3750	Y	MSM applications processor
S6	SMPS	0.735	0.4-1.14	0.350-1.355	3750	Y	MSM applications processor
S7	SMPS	0.750	0.4-1.14	0.33-2.04	2000	Y	MSM VDD memory rail (VDDMX)
L1	NMOS LDO	1.0 ²	1.0 ²	0.375-1.5375	600	N	RFICs
L2	NMOS LDO	1.2	1.2	0.375-1.5375	1200	Y	VREF generator for LPDDR3
L3	NMOS LDO	1.2	1.2	0.375-1.5375	600	N	Camera digital
L4	PMOS LDO	1.800	1.800	1.750-3.3375	450	N	RFICs and GPS eLNA
L5 ⁴	PMOS LDO	1.800	1.800	1.750-3.3375	600	Y	Most digital ICs, MSM pad groups 3 and 7, LPDDR, and eMMC
L6	PMOS LDO	1.800	1.800	1.750-3.3375	300	N	MSM GPRXCM, camera, touchscreen, display, and sensors
L7	PMOS LDO	1.800	1.800	1.750-3.3375	300	Y	MSM analog, USB and PLLs, WCN XO, and PM baseband clock driver
L8	PMOS LDO	2.900	2.900	1.750-3.3375	600	Y	eMMC
L9	PMOS LDO	$V_{bat} = 3.3 \text{ V}$ for $V_{bat} > 3.575 \text{ V}$ $V_{bat} = 3 \text{ V}$ for $V_{bat} < 3.575 \text{ V}$	3.000-3.300	1.750-3.3375	600	N	WCN
L10	PMOS LDO	3.0	3.0	1.750-3.3375	150	N	Sensors and touchscreen
L11 ⁵	PMOS LDO	2.900	2.900	1.750-3.3375	800	Y	Micro SD

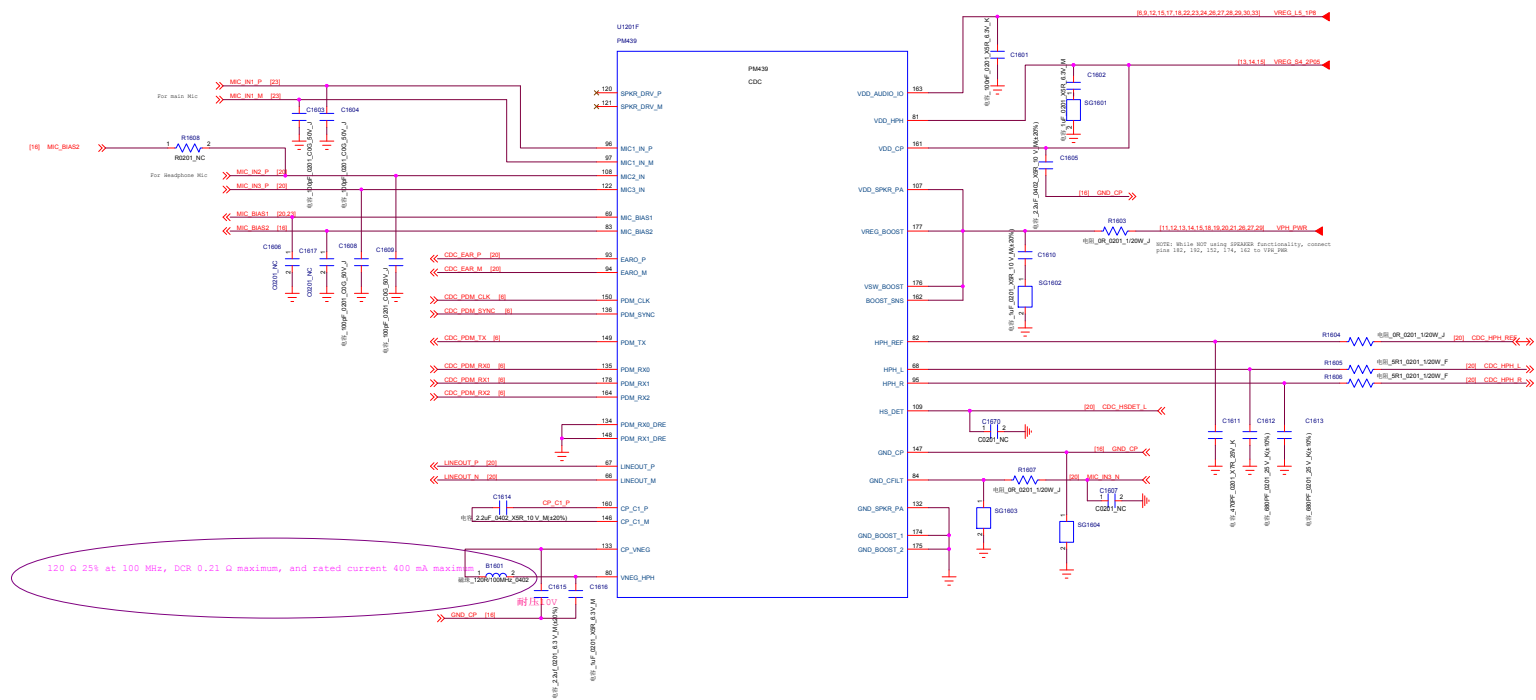
Table 3-6 PM439 regulators and their intended uses for SOM439/SOM429 when paired with PM632 (cont.)

Function	Circuit type	Default voltage (V) ¹	Specified range (V) ¹ (SOM439/SOM429)	Programmable range (V)	Rated current (mA)	Default on	Expected use (SOM439/SOM429)
L12 ⁴	PMOS LDO	2.950	1.800-2.950	1.750-3.3375	50	Y	MSM pad group 2
L13	PMOS LDO	3.075	3.075	1.750-3.3375	150	Y	MSM USB and PMIC and external codec audio
L14 ⁵	PMOS LDO	1.800	1.800	1.750-3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15 ⁵	PMOS LDO	1.800	1.800	1.750-3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750-3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750-3.3375	300	N	Camera and display
L18	PMOS LDO	2.700	2.700	1.750-3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375-1.5375	600	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.7625	1.7625	1.74-3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.7625	1.7625	1.74-3.3375	5	N	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750-3.3375	150	N	Camera: analog
L23	NMOS LDO	0.8	0.8	0.375-1.5375	600	Y	MPI CSI, DSI, and USB

- All regulators have default voltage settings, whether or not they default on; the voltage and state depends upon the programmable boot sequencer (PBS) configuration.
- The specified voltage range is the programmed range for which performance is guaranteed to meet all specifications. For usage outside this range, submit a case to QTI for approval.
- NOTE: LDO-related current specifications are only valid while maintaining their specified headroom.
- LD01 DC accuracy will be 2% at trimmed voltage (1.05 V) and 3% for other settings. Software configures this LD01 to 1 V, hence 3% DC accuracy must be factored in.
- L5 powers internal circuits and limited to 1.8 V operation; its programmed voltage should not be changed, and it should not be turned off.
- L11, L14, and L15, as well as all PMOS LDOs have OCP.

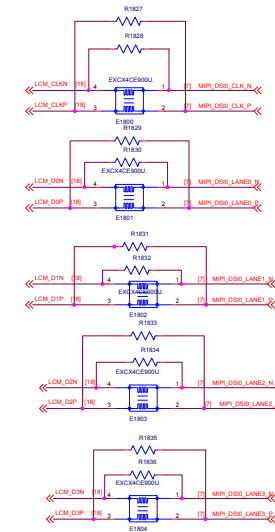
PM439 LDO

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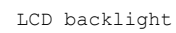


PM439 CODEC/Audio PA

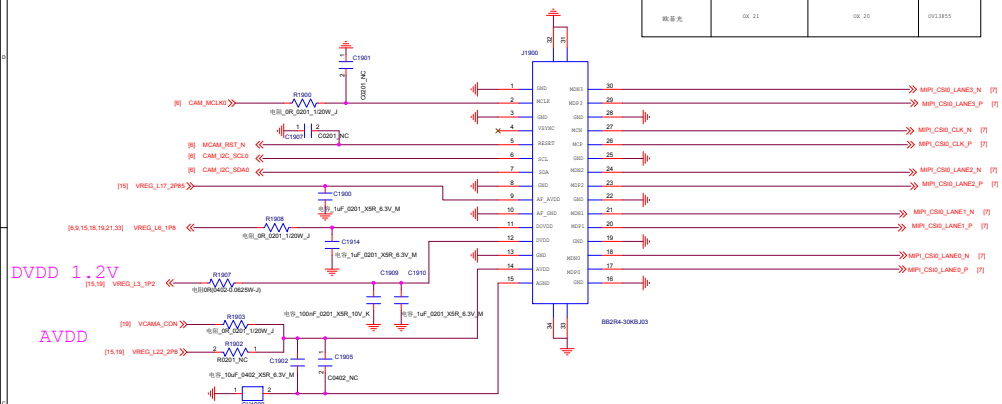
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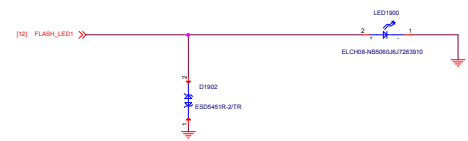
LCD-BACKLIGHT



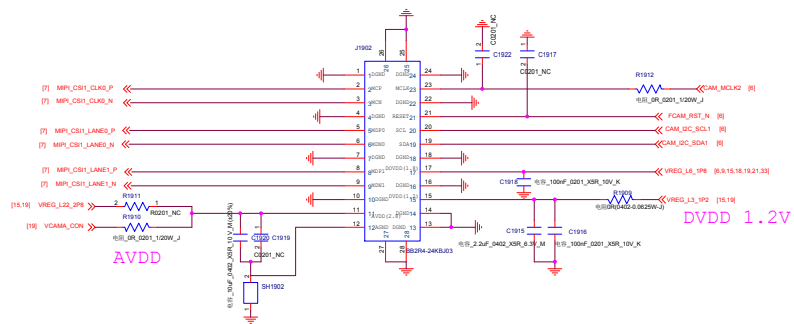
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品牌商	OK 21	OK 20	DFI3855
制造商	OK 21	OK 20	DFI3855



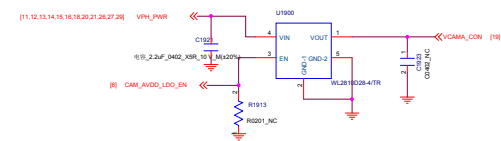
厂家	IPC 8640	IPC 8617R	BRIDGES DC
后铁圈	OK TA	OK TB	OV22A12-GBAA
欧普光	OK TA	OK TB	OV22A12-GBAA



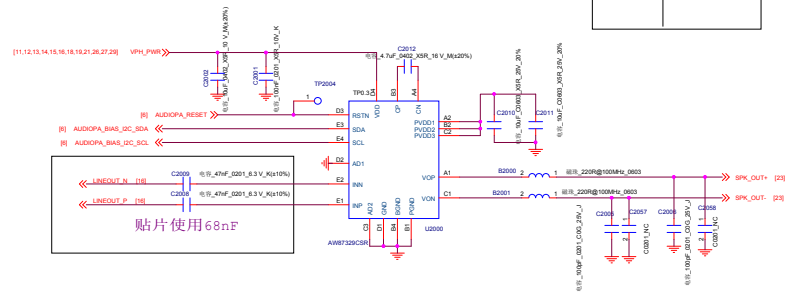
厂家	12C READ	12C WRITE	ID
后铁盘	OK 5A	OK 5B	3504871032-P025
欧普亮	OK 5A	OK 5B	3504871032-P025



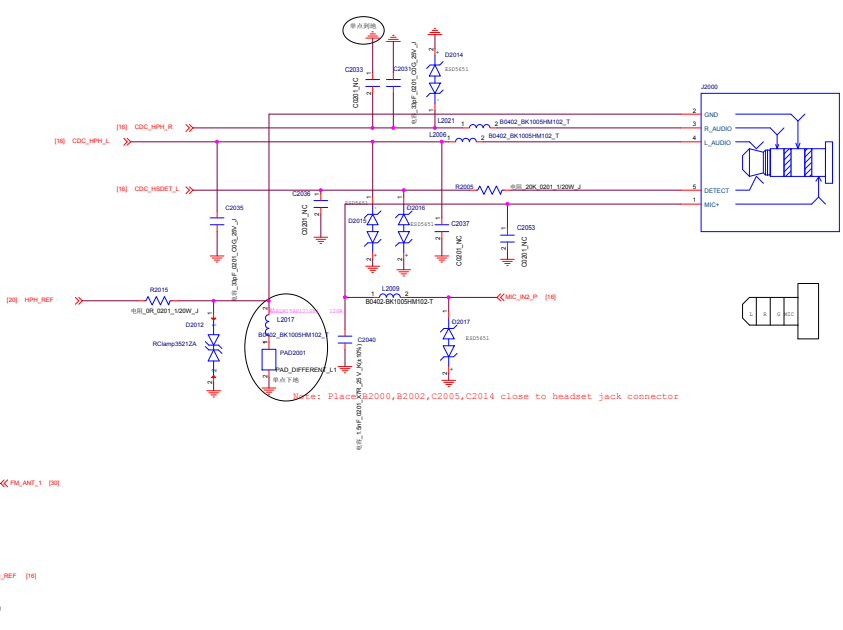
贴片使用WL2820D28-4/TR DFN1x1-4L



厂家	IIC ADDRESS
AM9732D	10110A2A1



American Standard EARPHONE



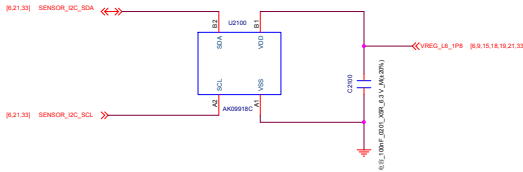
The image shows a detailed PCB layout for a USB-to-serial converter. Key components include:

- Connectors:** USB Type-A (X1), USB Type-B (X2), and DB9 (X3).
- ICs:** CH340 (U1) for USB-to-serial conversion, and a USB-to-UART bridge (U2, MAX3232CPE).
- Passive Components:** Capacitors (C1-C10), resistors (R1-R10), and inductors (L1-L5) for signal conditioning and decoupling.
- Layout Features:** A red circle highlights a specific area on the right side of the board, likely indicating a design note or a specific component placement.

Title			
Area_MB_V1			
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D	<Doc>	V1.0	
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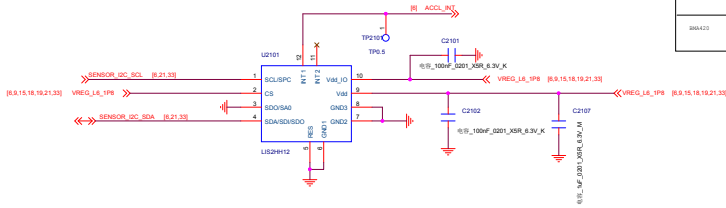
E-compass

厂家	12C A009025
A00918C	07h



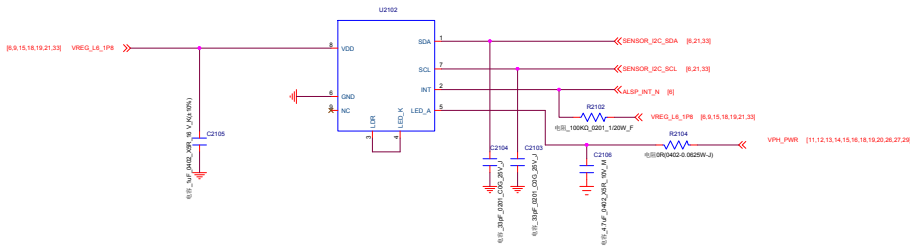
A sensor

厂家	12C A009025
A00918C	07h

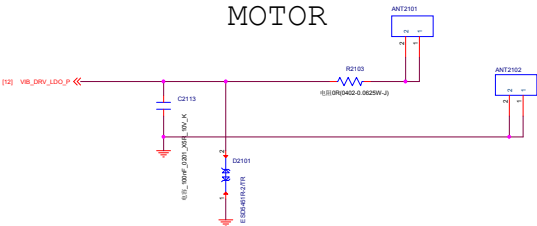


ALS PS

厂家	12C A009025
A00918C	07h



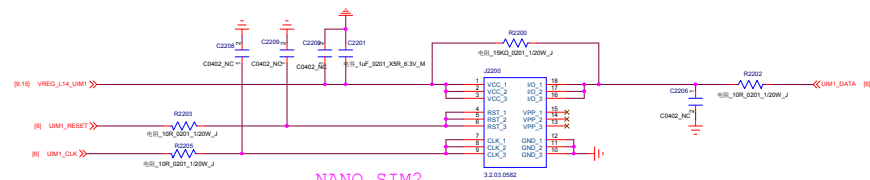
MOTOR



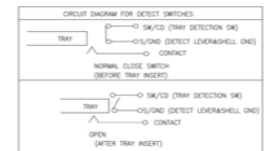
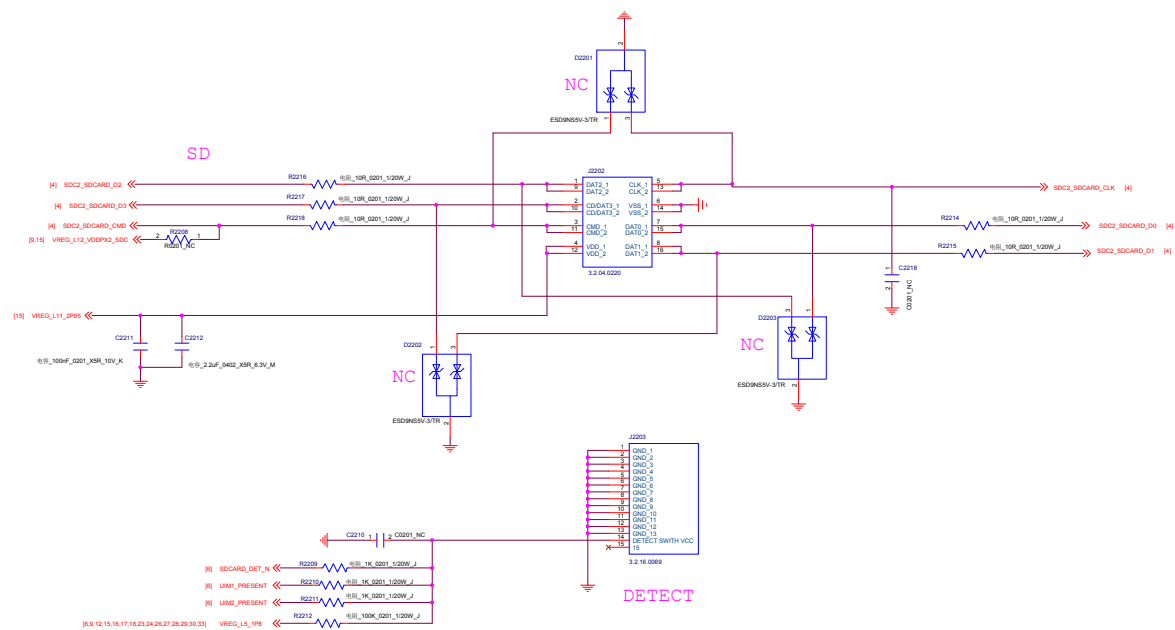
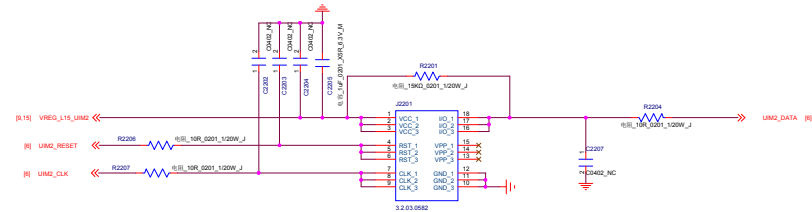
LED



NANO SIM1

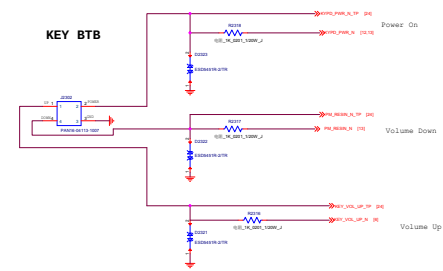
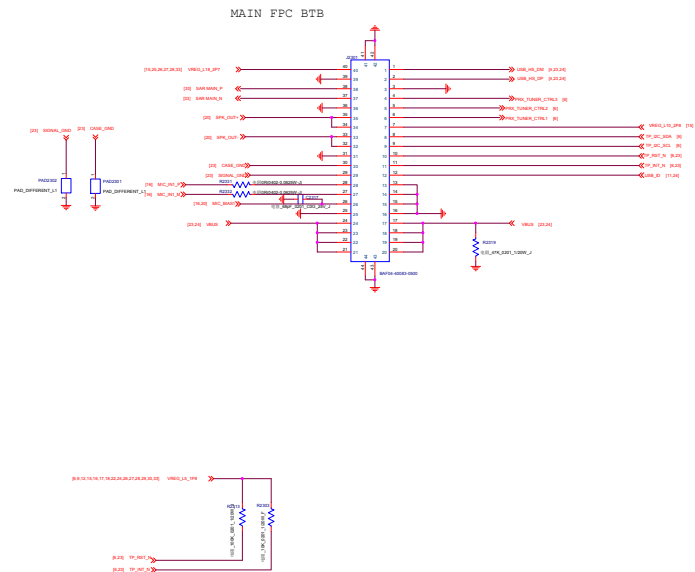
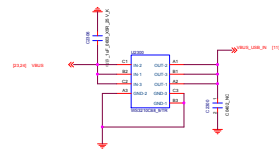


NANO SIM2

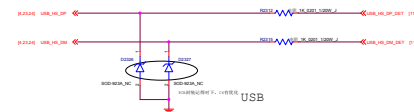
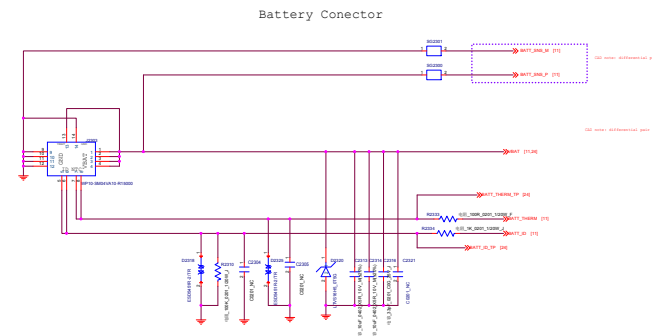


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OVP 6.8V WS3210C68-9/TR



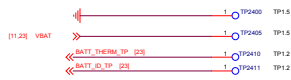
Signal	Description
KEY_VOL_UP_N	Volume Up
PM_RESIN_N	Volume Down
KYPD_PWR_N	PWER ON
KEY_AI	XIAOMI AI



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No.	Document Number
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Date:	Friday, March 26, 2010 12:07 PM

Test Points

VBAT TEST POINT



KEY TEST point



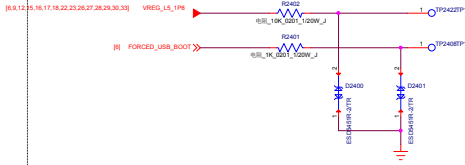
UART TEST point



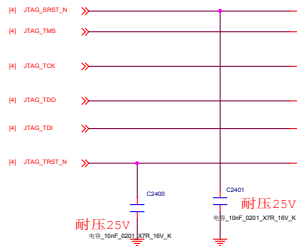
USB TEST POINT

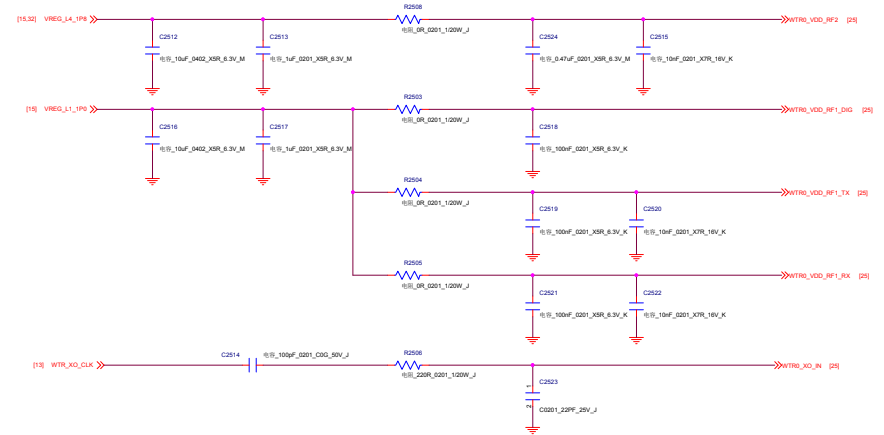
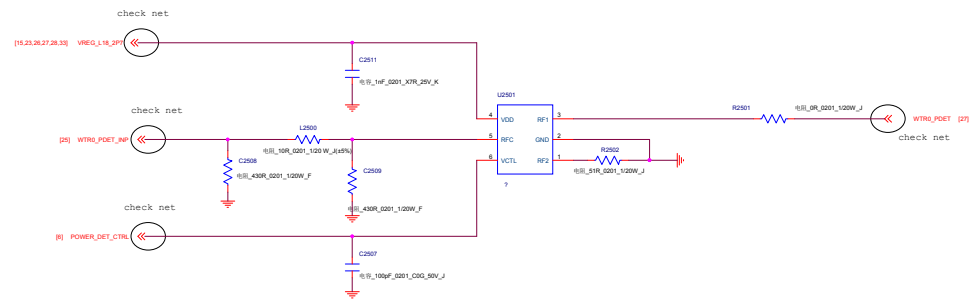


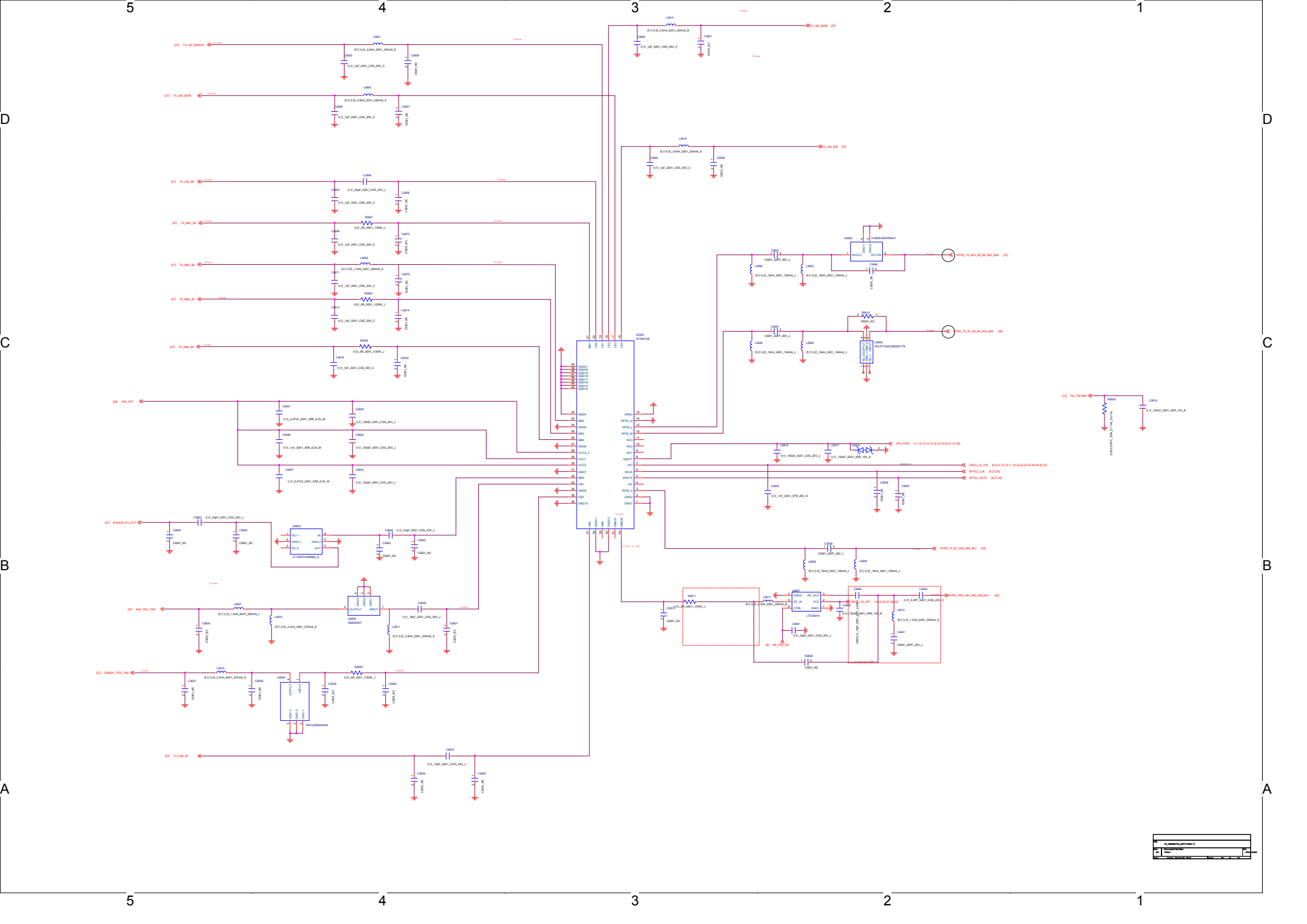
Force USB boot



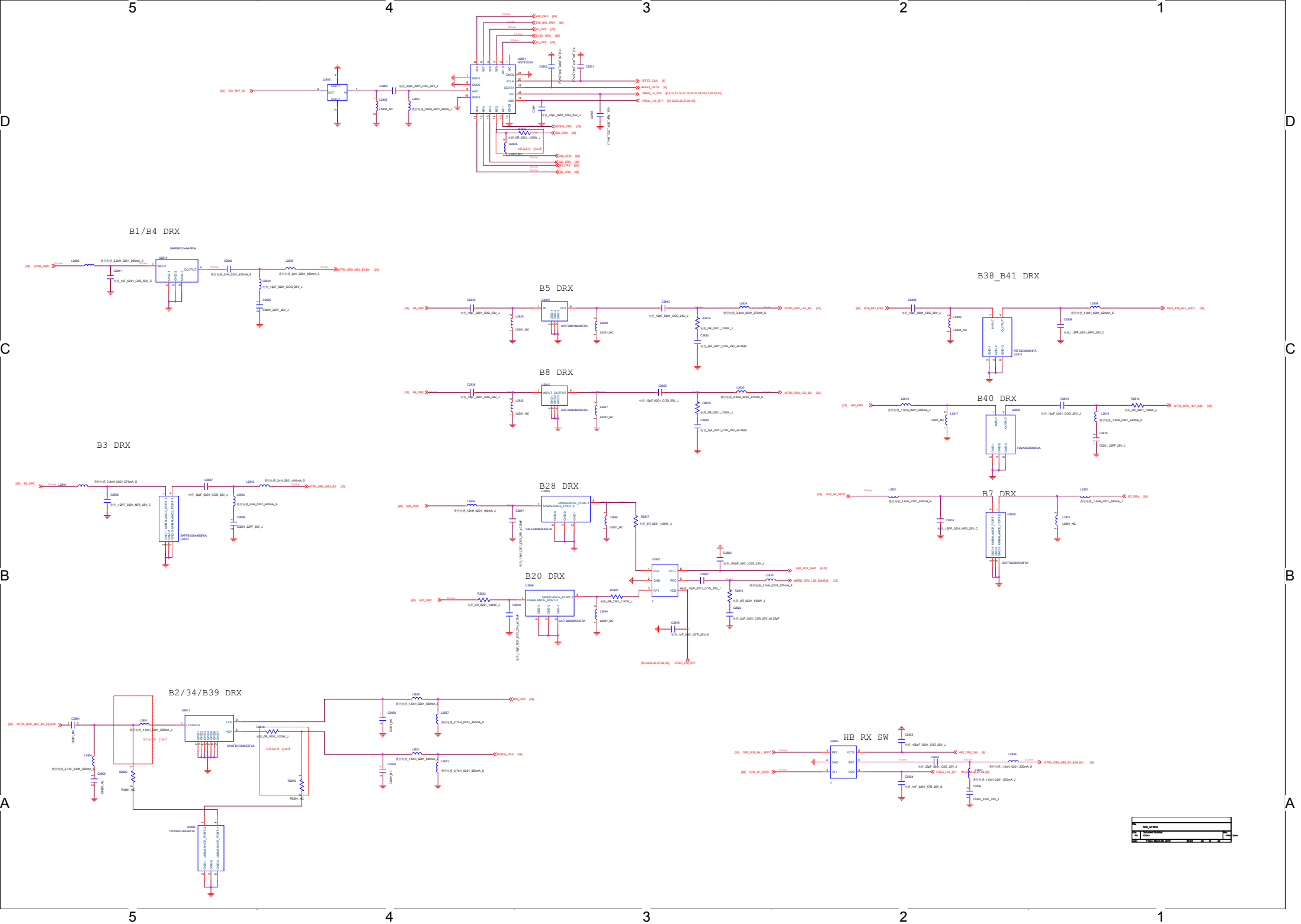
JTAG TEST POINT

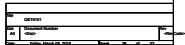


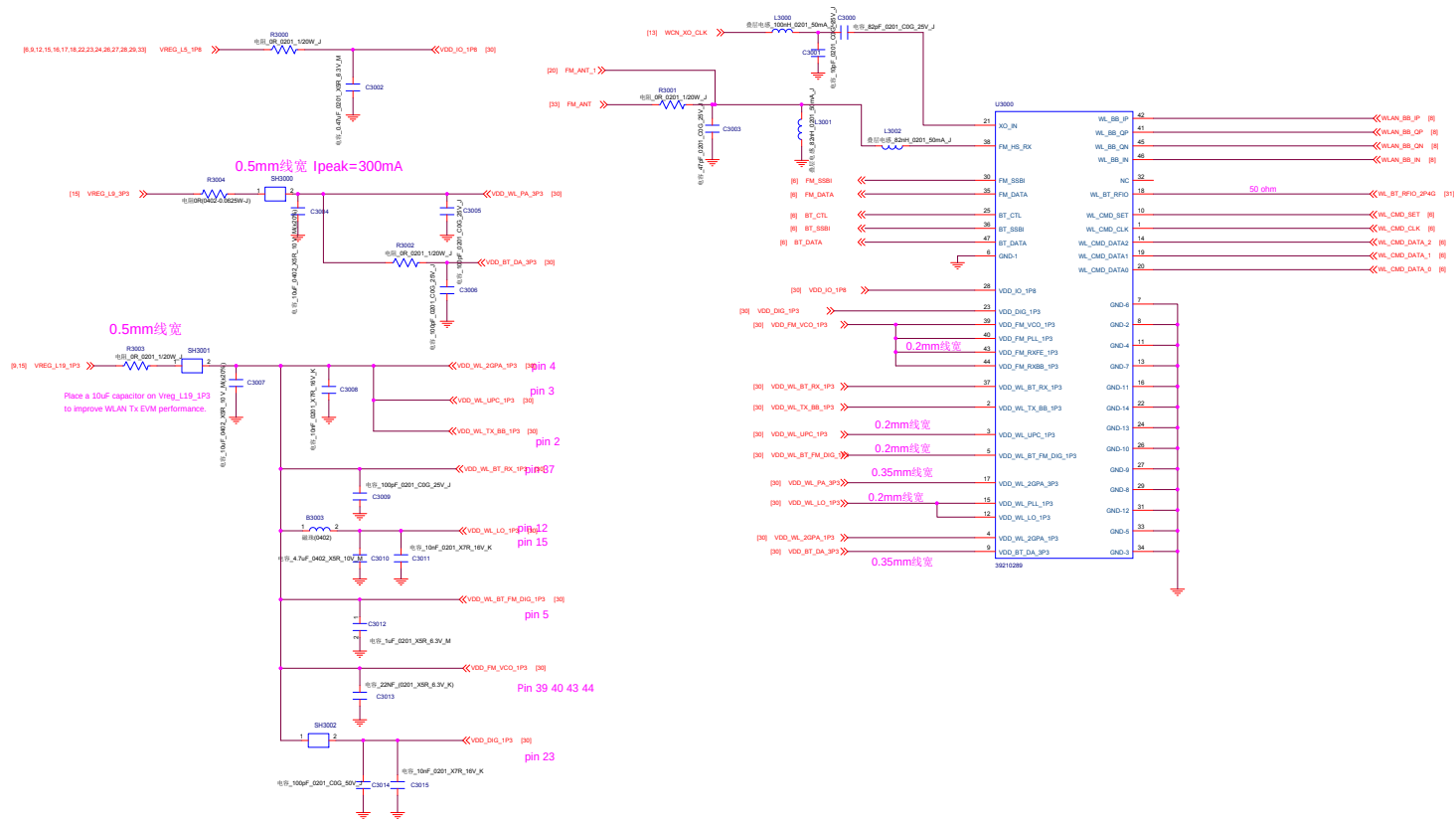


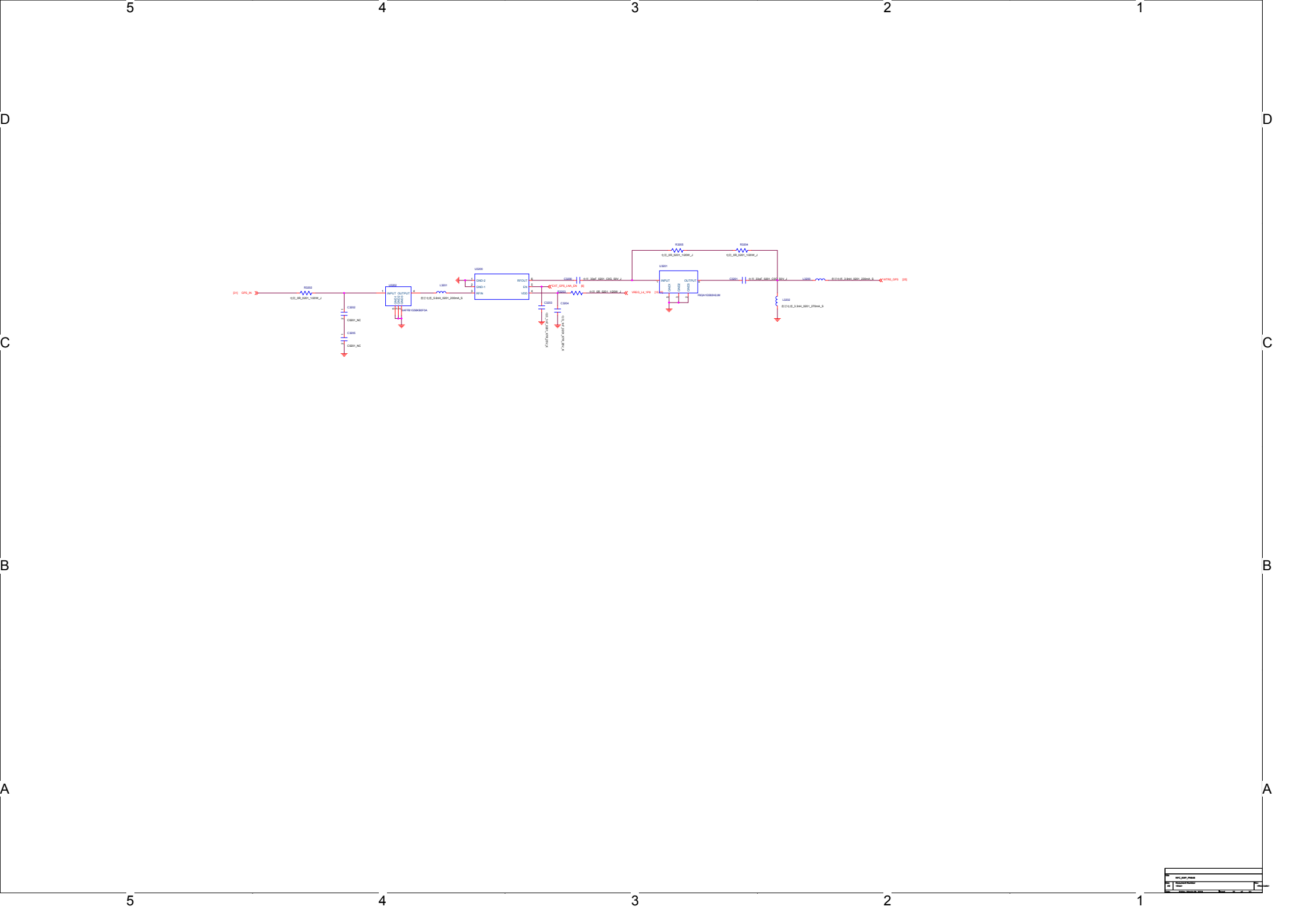




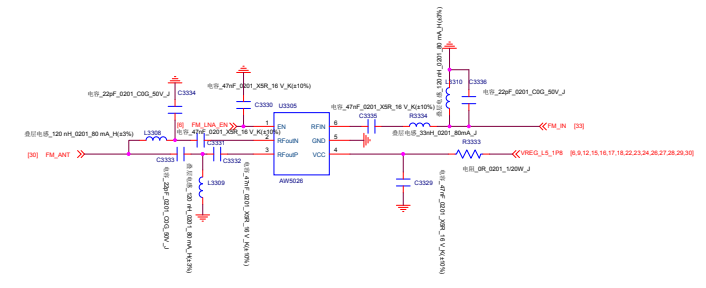








The diagram illustrates the antenna matching network for the R3309 and R3310 modules. The R3309 module (left) contains capacitors C3301, C3302, and C3303, and inductors R3309 and R3310. The R3310 module (right) contains capacitors C3304 and C3305, and inductors R3311 and R3312. Both modules are connected to a common 'share pad' which is then connected to the antenna input and output lines. The diagram also shows the connection to the antenna input and output lines, labeled as 'ANTENNA_IN' and 'ANTENNA_OUT'.

[illegible]

The schematic diagram illustrates the DIV ANT circuit. Key components include the RF1824 IC, capacitors C3248, C3344, C3345, C3326, C3337, and C3336, and inductors L3202, L3201, L3217, L3326, and L3336. The circuit is powered by a 1.2V supply (VDD_12V_3P1) and a 1.2V supply (VDD_12V_3P1). The output is connected to a 15-pin connector (VHS0_15_3P1). The circuit is labeled with various component values and connections, including a 15-pin connector (VHS0_15_3P1) and a 15-pin connector (VHS0_15_3P1).

[illegible]

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Title Schematic, CCA, MAIN, QRD438, 2-4-2, EV1.0	Rev A			